

**B.Tech. Degree VI Semester Regular/Supplementary Examination
May 2024**

**CE 19-201-0604 GEOTECHNICAL ENGINEERING II
(2019 Scheme)**

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Determine the earth pressure on retaining structures and apply the knowledge on the design of sheet pile walls.
 CO2: Apply the knowledge on soil exploration methods to carry out soil investigation for any civil engineering construction.
 CO3: Estimate the bearing capacity of soils by different methods.
 CO4: Estimate the probable settlements beneath foundations.
 CO5: Estimate pile and pile group capacity for any type of soil.
 CO6: Achieve knowledge of design and construction of well foundations and cofferdams.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer **ALL** questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Illustrate about the at-rest stress condition of a retaining wall.	3	L1	1	1
(b) Discuss about different types of sheet pile walls.	3	L3	1	2
(c) Define a bore log.	3	L2	2	5
(d) List out the different types of samplers used as part of soil exploration.	3	L4	2	1
(e) Illustrate the Terzaghi's classification of shallow and deep foundations.	3	L1	3	3
(f) What do you understand by uniform and differential settlement?	3	L3	4	2
(g) Write down the dynamic formulae for the determination of pile capacity.	3	L3	5	7
(h) Enlist and sketch the different shapes of well foundations.	3	L3	6	1

PART B

(4 × 12 = 48)

- II. (a) Outline the steps for the construction of Culmann line and illuminate the procedure for determining the active earth pressure on a retaining wall having an inclined backside and inclined backfill ground surface. 6 L5 1 3
- (b) A retaining wall 6 m high with vertical back supports clay as backfill with a horizontal surface. The properties of backfill are $\Phi = 0^\circ$, $C = 35 \text{ kN/m}^2$ and $\gamma = 17 \text{ kN/m}^3$. Determine the following: 6 L5 1 2
- Depth of tension crack
 - Critical depth
 - Total active thrust after the formation of tension crack and its point of application.

OR

- III. (a) Discuss the analysis of a free earth support cantilever sheet pile in cohesive soil. 6 L4 1 2
- (b) Deliberate the variation of coefficient of lateral earth pressure with reference to the stress condition. 6 L2 1 3

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Describe the different stages of a site investigation programme.	6	L6	2	1
	(b) Brief out regarding the specifications of SPT.	6	L4	2	4
OR					
V.	(a) Explain about rotary drilling.	6	L5	2	4
	(b) Discuss the utility of geophysical methods for site investigation.	6	L1	2	1
VI.	(a) Determine the ultimate bearing capacity for a square footing having size $2\text{ m} \times 2\text{ m}$ located at a depth of 1.5 m in stiff clay of unit weight 20 kN/m^3 . The shear strength parameters of the clay at a depth of 1.5 m are $c = 130\text{ kN/m}^2$ and $\phi = 0$.	6	L5	3	3
	(b) Elaborate the Skempton's analysis for cohesive soils.	6	L2	3	1
OR					
VII.	(a) Estimate the immediate settlement of a concrete footing $1.5\text{ m} \times 1.5\text{ m}$ in size founded at a depth of 1 m in silty soil having modulus of elasticity 9000 kN/m^2 . The footing is expected to transfer a unit pressure of 200 kN/m^2 . Take $\mu = 0.35$ and $I_f = 0.82$.	6	L6	4	2
	(b) Discuss the applicability of a mat foundation.	6	L3	3	1
VIII.	(a) A precast concrete pile of diameter 30 cm is driven into homogeneous clay stratum. The unconfined compressive strength of the clay is 60 kN/m^2 . Determine the length of pile required to carry a safe working load of 40 kN with a factor of safety of 3. (Take $\alpha = 0.5$).	6	L5	5	3
	(b) Discuss about the advantages and limitations of driven piles.	6	L3	5	1
OR					
IX.	(a) A group of 4 piles with 2 piles in a row was driven into soft clay extending from ground level to a great depth. The diameter and length of piles were 40 cm and 9 m respectively. The cohesion of clay is 25 kN/m^2 . If the centre to centre spacing of the piles were 70 cm , determine the allowable load on the pile group for a factor of safety of 3. (Take $\alpha = 1$).	6	L5	5	3
	(b) Discuss in detail about the remedial measures adopted for rectifying the tilts and shifts of a well foundation.	6	L2	6	1

Blooms's Taxonomy Level's

L1 - 10%, L2 - 17.5%, L3 - 20%, L4 - 12.5%, L5 - 30%, L6 - 10%.
